

TIC-TAC-TOE SYMBOL RECOGNITION

1. Overview

The dataset is designed for a multi-class image classification task within the domain of Computer Vision. The primary objective is to identify and categorize individual cells of a Tic-Tac-Toe board from image data. This dataset serves as the foundational input for the `TicTacToeCNN` architecture.

2. Class Distribution

The data is perfectly balanced across three categories to ensure the model does not develop a bias toward any specific symbol.

Category	Description	Train Count	Test Count	Total
Blank	Empty board cell	160	40	200
Circle	'O' Player symbol	160	40	200
Nought	'X' Player symbol	160	40	200
Total		480	120	600

3. Technical Specifications

The images are stored in a format optimized for deep learning pipelines:

- Color Space:** Grayscale (Single Channel).
- Dimensions:** 128 * 128 pixels.
- Format:** Optimized for `torchvision.datasets.ImageFolder` compatibility.
- Variability:** Contains hand-drawn style variations, different line weights, and center-offset symbols to simulate real-world game conditions.